

Beyond the Event Horizon:

Persistent Operator Structure at Rigidity Walls — from Four-Body Shape Space to Black-Hole Ringdown

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Abstract

Black-hole ringdown is governed by non-normal operators: quasinormal modes are eigenmodes of a non-selfadjoint Hamiltonian, their spectra are unstable in the pseudospectral sense, and their non-orthogonality produces transient plateaus and transient superradiance. Those statements are established results of the modern literature [1, 2, 3, 5, 6, 7], not claims of this paper. What this paper contributes is the other side of a correspondence: the same non-normal phenomenology arises, with exact and independently gated machinery, in the relational shape space of the four-body problem. We derive, from the quotient construction alone, the closed-form kinetic metric of the five-dimensional shape space Σ_4 — and find it is exactly the Bures–Wasserstein metric of trace-one positive matrices, so that a four-body system’s shape manifold is isometric to the (real) state space of a qutrit, its collision stratification the rank stratification of density matrices. We derive the Jacobi curvature of Σ_4 by the O’Neill submersion formula, validated against the Kendall shape sphere ($K = 4.0000000$), and show it is *finite* at interior eigenvalue coincidences — withdrawing, with the computation on record, the divergent shear entries of an earlier draft — while three derived mechanisms co-locate on the genuine codimension-one boundary, the coplanar wall $w_3 = 0$: the metric blows up as exactly $1/(6w_3)$, the curvature as C/w_3 , and the transport operator forms exact size-2 Jordan blocks there under an explicit joint condition. The horizon interface of the correspondence is staged on that wall. Two survival principles organize what remains observable through the degeneracy, each demonstrated by gated computation: when eigenvectors fail, the *phase* carries the record ($\|v_\perp\| \propto 1/\sqrt{2\lambda} \rightarrow 0$ while $\int \sqrt{2\lambda} dt$ persists); when eigenvalues fail, the *Dunford contour* carries it ($\text{Tr } \Pi = 2.000000000000$ and the trace-log ledger exact to 6×10^{-15} at the exactly defective point, where rank-one data diverges by the snapped law $\|P_1\| \cdot \text{gap} \rightarrow |A_{12}|$). The correspondence with black-hole horizons is then stated at its honest grade: a theorem about shared block structure on the shape side, citations on the general-relativity side, and a clearly labeled conjecture for the dictionary. All scripts are included in the paper as the proof of record.

0 What is claimed, and what is not

The premise that horizon physics is non-normal operator theory is *not* claimed here; it is the settled starting point of an active program. Quasinormal modes cast on hyperboloidal slices are the spectral problem of a non-selfadjoint operator whose spectrum is unstable under high-frequency perturbations [1, 3]; the instability has gravitational-wave signatures [2]; the choice of scalar product is structural, not cosmetic [4]; non-orthogonal quasinormal sums produce arbitrarily long-lived transient plateaus — localized energy packets near the future horizon — and transient growth in higher-derivative Sobolev norms [5, 7]; and transient superradiant amplification occurs in modally stable regimes [6]. Sections 5 and 8 of this paper lean on those results and attribute them.

What is claimed is the *relational side* and the bridge. First, exact geometry: the four-body shape space carries, by its own quotient construction, the Bures–Wasserstein metric (Theorem 1), identifying it with quantum state space and its collision walls with rank strata. Second, exact boundary structure: metric blow-up, curvature blow-up, and Jordan-block formation are derived and co-located on the coplanar wall (Section 3). Third, an organizing pair of survival principles — phase and contour — each demonstrated by computation on the audited operator (Section 4). Fourth, a correspondence with horizon interfaces stated as one theorem, one dictionary, and one conjecture, in that order and at those grades (Section 6).

Three grades of statement. As in the companion spectral-stress work, every claim carries one of three grades, never silently promoted: *exact/derived* (closed forms and identities with gated scripts behind them); *measured* (numerical constants quoted with their depths); and *proposed* (ansätze and conjectures, fenced in Sections 6 and 8). Where a chain mixes grades, the weakest link is named.

Two survival principles. The paper’s method can be said in one sentence each. *When you lose sight of the vectors, the phase keeps the trace:* tied or oscillatory modes defeat eigenvector tracking, but a phase rate remains a well-defined invariant — the mechanism behind the calibrated phase-rate result of [24] and behind the wall oscillator of Section 3. *Dunford calculus keeps it wrapped up:* when eigenvalues coalesce, contour-integrated functional calculus — Riesz projectors, trace-logs — remains exact where eigenvector arithmetic has no meaning (Section 4). Both principles are demonstrated, not asserted, and the demonstrations are in Appendix B.

1 The shape space and its true geometry

Let a four-body configuration in $\mathbb{R}^3$ be encoded, after centering and mass-weighting, by the 3×3 matrix X of Jacobi vectors. Rotations act on the left, $X \mapsto RX$, and the rotation-invariant datum is the Gram matrix $G = X^\top X$ — not $XX^\top$, which transforms as $RM R^\top$. Fixing overall scale by $\text{Tr } G = 1$ (equivalently $\|X\|_F = 1$), the shape space is

$$\Sigma_4 = \{ G \succeq 0 \text{ symmetric, } \text{Tr } G = 1 \},$$

five-dimensional, stratified by rank. In the principal frame $W = \text{diag}(w_1, w_2, w_3)$, $\sum w_i = 1$, the tangent space is spanned by the Frobenius-orthonormal basis

$$E_1 = \frac{1}{\sqrt{2}} \text{diag}(1, -1, 0), \quad E_2 = \frac{1}{\sqrt{6}} \text{diag}(1, 1, -2), \quad E_3, E_4, E_5 = \frac{1}{\sqrt{2}} (e_i e_j^\top + e_j e_i^\top)_{(12), (13), (23)}.$$

E_1, E_2 deform the eigenvalues (shape); E_3, E_4, E_5 rotate the eigenframe (shear).

The physical metric on Σ_4 is not a choice: it is the quotient of the flat Frobenius metric on the unit sphere $\|X\|_F = 1$ by the rotation action. Computing the horizontal lifts explicitly (script BH-CURV-01, Appendix B) yields closed forms.

Theorem 1 (the quotient metric, in closed form). *In the principal frame, the kinetic (submersion) metric on Σ_4 is*

$$g(E_a, E_b) \Big|_{\text{shape}} = \sum_{i=1}^3 \frac{(E_a)_{ii} (E_b)_{ii}}{4 w_i}, \quad g(E_{(ij)}, E_{(ij)}) \Big|_{\text{shear}} = \frac{1}{2(w_i + w_j)},$$

with the shape and shear sectors orthogonal and distinct shear directions mutually orthogonal.

Derivation and verification. The horizontal lift H of a base direction E at $X = \text{diag}(\sqrt{w_i})$ solves the linear system $X^\top H + H^\top X = E$, $XH^\top$ symmetric; the metric is $g(E_a, E_b) = \langle H_a, H_b \rangle_F$. For shape directions H is diagonal with $h_i = (E)_{ii}/(2\sqrt{w_i})$, giving the first formula; for the (ij) shear direction, $H_{ij} = \sqrt{w_i}/(\sqrt{2}(w_i + w_j))$, $H_{ji} = \sqrt{w_j}/(\sqrt{2}(w_i + w_j))$, giving the second. Verified numerically at generic points: $g_{33} = 0.625 = 1/(2 \cdot 0.8)$, $g_{44} = 0.714286 = 1/(2 \cdot 0.7)$, $g_{55} = 1.000 = 1/(2 \cdot 0.5)$ at $w = (0.5, 0.3, 0.2)$, and the shape block to machine precision. $\square$

The identification. These are the Bures weights. The quotient of the Frobenius metric on square matrices by the orthogonal action, transported along $X \mapsto X^\top X$, is the Bures–Wasserstein (equivalently Procrustes) geometry of positive semidefinite matrices [20, 21]; restricted to trace one it is the Bures geometry of density matrices [19, 22], the quantum-Fisher (Helstrom) metric of quantum information. Hence:

The four-body shape space with its own kinetic metric is isometric to the state space of a real qutrit. Its collision stratification is the rank stratification of density matrices: the coplanar wall is the rank-2 stratum, the collinear stratum is rank 1.

The geometry is classical [20, 21, 22]; the application to relational N -body mechanics — and everything built on it below — is this paper’s. The identification also situates the three-body shape sphere: at $d = 2$ the same construction gives the Bures geometry of real qubits, whose pure-state boundary is the round sphere of radius $\frac{1}{2}$ — the Kendall shape sphere — which is the control used to validate the curvature machinery next.

2 The stability spine, done right

The covariant evolution of a horizontal deviation field J along a shape trajectory obeys the reduced Jacobi equation

$$\nabla_t^2 J + R(J, \dot{W})\dot{W} + \nabla^2 U(W) \cdot J = 0, \quad (1)$$

with first-order form $\nabla_t Y = A_{\text{Jacobi}} Y$, $A_{\text{Jacobi}} = \begin{pmatrix} 0 & I_5 \\ -K & 0 \end{pmatrix}$, $K = K_{\text{curv}} + \text{Hess } U$. This section supplies the two ingredients of K at the exact/derived grade, and withdraws what an earlier draft printed in their place.

2.1 The transport block structure

The intrinsic transport operator of the series, with the trace-removal coefficient corrected to preserve tangency, is $A(H) = TH + HT - 2 \text{Tr}(TH) W$ with $T = \text{diag}(t_i)$ the mass-inertia data. Its verified block structure in the basis of Section 1 (scripts FOURBODY-AUDIT-01/02/03):

$$\begin{aligned} A_{11} &= (t_1+t_2) - (t_1-t_2)(w_1-w_2), & A_{22} &= 2t_3 + w_3(t_1+t_2-2t_3), \\ A_{12} &= \frac{1}{\sqrt{3}}[(t_1-t_2) - (t_1+t_2-2t_3)(w_1-w_2)], & A_{21} &= \sqrt{3}(t_1-t_2)w_3, \end{aligned} \quad (2)$$

the shear block *exactly diagonal* with entries t_1+t_2 , t_1+t_3 , t_2+t_3 , and the shape–shear coupling *identically zero* in the principal frame. Non-normality requires both anisotropies: $\|[A, A^\top]\| = 1.78$ with T and W both anisotropic, 6×10^{-16} with W isotropic, 0 with T isotropic; its source is the trace-removal term, and any expression of the form $\sum_{i<j} (t_i - t_j)^2 f(w)$ must carry $f(W \propto I) = 0$.

A withdrawal. *An earlier draft printed shear-block entries carrying $(\dot{w}_i - \dot{w}_j)^2/(w_i - w_j)^2$ and velocity-dependent shape-shear couplings. Those entries are withdrawn, on two independent computations. For the transport operator, the audit shows the shear block is diagonal with no w -dependence and the coupling vanishes under the paper’s own principal-frame assumption. For the Jacobi curvature, the fresh derivation below shows the curvature is finite at interior eigenvalue coincidences $w_i = w_j$: such coincidences are not orbit-type strata of the quotient (the stabilizer of a full-rank X is trivial regardless of the spectrum of G), so nothing singular lives there. The divergences arose from treating first-order eigenvector perturbation theory, $\Omega_{ij} = \dot{W}_{ij}/(w_i - w_j)$, as a mechanical connection. The conclusion those entries served — that the boundary is a rigidity interface — survives and is now derived, at the correct stratum, in Section 3.*

2.2 The curvature, derived

Since no derivation of K_{curv} existed anywhere in the series, it is supplied here from the construction alone. The total space is the unit sphere $\|X\|_F = 1$ (curvature 1), the quotient is by the isometric rotation action, and the O’Neill formula gives the base curvature exactly:

$$\langle R(J, V)V, J' \rangle = \underbrace{g(V, V)g(J, J') - g(J, V)g(V, J')}_{\text{sphere}} + 3\langle A_V \tilde{J}, A_V \tilde{J}' \rangle, \quad (3)$$

where $A_V \tilde{J}$ is the vertical part of the derivative of the horizontal-lift field — linear algebra at a point, no model input. Script BH-CURV-01 implements this with four gates: (i) *Kendall control*: at $d = 2$ the code returns the shape sphere’s sectional curvature as 4.0000000 at three distinct points; (ii) symmetry of the quadratic form, exact; (iii) $R(V, \cdot)V$ annihilates V to 4×10^{-11} ; (iv) the closed-form metric of Theorem 1 recovered. Two structural facts follow at the derived/measured grade:

- **Interior finiteness.** As $w_1 \rightarrow w_2$ at fixed $w_3 > 0$, every curvature entry converges (e.g. $K_{33} \rightarrow 1.855469$): the quotient geometry is smooth across interior eigenvalue crossings, consistent with the orbit-type analysis and with the general nonnegative-curvature theory of the Bures–Wasserstein quotient [21].
- **Wall divergence.** As $w_3 \rightarrow 0$ the curvature diverges at first order: $K_{44} w_3 \rightarrow 0.0713$, $K_{55} w_3 \rightarrow 0.1033$, $K_{22} w_3 \rightarrow 0.701$ (measured over three decades).

3 The wall

The stratification must be stated correctly before any boundary physics is staged on it. Within Σ_4 (dimension 5): the coplanar stratum $w_3 = 0$ (rank 2) has codimension *one* — it is the only genuine wall, and it is the branch locus over which the oriented shape space double-covers the reflection shape space; the collinear stratum $w_2 = w_3 = 0$ (rank 1) has codimension 3; and binary collisions B_{ij} are codimension-3 subsets of the rank-degenerate locus — generic trajectories miss them entirely. An earlier draft staged its horizon analysis at B_{ij} ; everything below relocates it to the coplanar wall, where three independent, derived mechanisms already live.

Theorem 2 (the rigidity interface, derived). *At the coplanar wall $w_3 \rightarrow 0$ of Σ_4 :*

1. **Metric blow-up (exact).** *The kinetic metric diverges in the E_2 direction as*

$$g(E_2, E_2) = \frac{1}{6w_3} (1 + O(w_3)),$$

directly from Theorem 1; measured ratio $g_{22} \cdot 6w_3 \rightarrow 1.000101$ at $w_3 = 10^{-4}$. Approaching the wall costs unboundedly in the kinetic metric: the “inertial squeeze” $\lambda \rightarrow \infty$ of the earlier draft is this statement, now derived rather than postulated.

2. **Curvature blow-up (derived law, measured constants).** The Jacobi curvature diverges as C/w_3 with the constants of Section 2.
3. **Jordan blocks (exact).** The transport shape block (2) becomes exactly triangular on the wall ($A_{21} = \sqrt{3}(t_1 - t_2)w_3 \rightarrow 0$, at rate $O(w_3)$), and under the joint condition

$$t_1 + t_2 - 2t_3 = (t_1 - t_2)(w_1 - w_2)$$

its eigenvalues coincide with $A_{12} \neq 0$: a genuine size-2 Jordan block (verified: normalized eigenvector determinant 1.7×10^{-15} with $A_{12} = 0.52$). Away from the wall the spectrum is real on the entire regular stratum — the discriminant of (2) is a quadratic form whose determinant is exactly $16w_1w_2w_3 \geq 0$ — so exceptional points exist only on the boundary strata.

The wall is thus a rigidity interface in three registers at once: metrically (motion into it is infinitely costly), curvature-wise (tidal stresses diverge), and spectrally (the transport operator degenerates into a Jordan block there and only there). The transverse first-order dynamics at the wall is the oscillator $\ddot{y}_\perp + 2\lambda y_\perp = 0$ with $\lambda \rightarrow \infty$: eigenvalues $\pm i\sqrt{2\lambda}$, velocity amplitude

$$\|v_\perp(t)\| \propto \frac{1}{\sqrt{2\lambda}} \rightarrow 0, \quad (4)$$

so incoming trajectories flatten and skim the wall — while the oscillator’s *phase* $\int \sqrt{2\lambda} dt$ remains finite-rate bookkeeping of the encounter. That is the first survival principle in action, and the honest content of the “haircut”: the transverse amplitude is sheared away; the phase record is not.

4 The two survival principles, demonstrated

4.1 Phase keeps the trace

When the spectral data ties or oscillates, eigenvector tracking fails while a phase rate remains invariant. This is a calibrated fact of the companion diagnostic work [24]: for a recurrence whose relevant modes are a tied complex pair, the tracked direction rotates indefinitely, and the surviving invariant is the pair’s beat rate — measured there against theory ($2\pi/3\sqrt{3}$) with the gap shrinking monotonically. On Σ_4 the same principle appears at the wall through (4): the vector datum $v_\perp$ vanishes; the phase $\int \sqrt{2\lambda} dt$ persists.

4.2 The contour keeps the ledger

When eigenvalues coalesce, rank-one spectral data diverges but contour-integrated (Dunford) data does not. Script **BH-LEDGER-01** demonstrates this on the audited operator itself, at the wall’s Jordan point ($t = (2, 1, 1.3)$, $w_1 - w_2 = 0.4$, $\lambda^* = 2.6$, $A_{12} = 0.485$), approaching along $w_3 = \varepsilon \rightarrow 0$:

ε	gap	$\ P_1\ $ (rank-one)	$\text{Tr } \Pi$ (contour)	ledger $-2 \log \lambda^*$
10^{-2}	1.8×10^{-1}	2.7	2.000000000000	-1.2×10^{-3}
10^{-4}	1.8×10^{-2}	26.5	2.000000000000	-1.2×10^{-5}
10^{-6}	1.8×10^{-3}	264.6	2.000000000000	-1.2×10^{-7}
0	0	∞	2.000000000000	-6.0×10^{-15}

Three facts, each gated. *The divergence law snaps closed:* $\|P_1\| \cdot \text{gap} \rightarrow |A_{12}|$ (0.485147 vs. 0.484974) — the blow-up constant of the rank-one projector *is* the Jordan coupling. *The contour survives the collision:* the Riesz projector $\Pi = \frac{1}{2\pi i} \oint_{\Gamma} (\lambda I - A)^{-1} d\lambda$ has $\text{Tr } \Pi = 2$ to twelve digits at every depth including the exactly defective point, with $\|\Pi\|$ flat. *The trace-log ledger is exact where eigenvectors are meaningless:* $\frac{1}{2\pi i} \oint_{\Gamma} \log \lambda \text{Tr } R(\lambda) d\lambda = 2 \log \lambda^*$ to 6×10^{-15} at $\varepsilon = 0$. One hypothesis is visible in the data: at $\varepsilon = 10^{-1}$ the pair sits outside the contour and the integral correctly reports nothing — the contour must enclose the colliding pair with the spectral-gap margin, which is exactly the hypothesis of the perturbation bounds in Appendix A.

The Hairdresser, as a named interpretation. With both principles in hand, the no-hair analogy can be stated honestly. The interface executes rank compression, not destruction: the *scissors* are the exact vanishing of the back-coupling ($A_{21} = O(w_3) \rightarrow 0$: block-triangular decoupling); the *fingers* are the transverse suppression (4) that flattens trajectories parallel to the wall; the *mirror* is the trace-log ledger, exact at the defective point; the *light* is the Dunford contour that keeps the enclosed invariants finite. This is imagery for derived structure — each phrase names a gated computation — and it is only imagery: no claim about information in quantum gravity is made.

5 Pseudospectral transients: attribution and the shape-space side

On the black-hole side, the transient story is the literature's. Horizons make the perturbation operator non-selfadjoint; quasinormal modes are non-orthogonal in physically relevant inner products; non-orthogonality yields transient plateaus — arbitrarily long-lived quasinormal sums, localized energy packets near the future horizon — and transient growth measured in higher-derivative Sobolev norms [5, 7]; the pseudospectrum organizes the spectral instability [1, 3]; and the choice of norm is a structural input [4]. Transient superradiance occurs in modally stable regimes [6].

On the shape-space side, the same structure follows from the machinery above. The stability tensor satisfies $[K, K^\top] \neq 0$ under the anisotropy conditions of Section 2, so eigenfunctions are non-orthogonal, $\langle \phi_n, \phi_m \rangle_{H^1(\Sigma_4)} = \Delta_{nm} \neq \delta_{nm}$, and a superposition of decaying modes holds its energy across a plateau whose duration is controlled by the Kreiss constant:

$$\|J(t)\|_{H^1(\Sigma_4)}^2 = \sum_n |c_n|^2 e^{2\text{Im}(\omega_n)t} + \sum_{n \neq m} c_n^* c_m e^{i(\omega_n^* - \omega_m)t} \Delta_{nm}, \quad \sup_t \|e^{-iA_{\text{Jacobi}}t}\| \geq K_C,$$

with $K_C = \sup_{\text{Im } \omega > 0} \text{Im}(\omega) \|(\omega I - A_{\text{Jacobi}})^{-1}\|$ (the Kreiss bound controls the supremum over time, not each instant). The inner product on $T\Sigma_4$ is not free: the quotient metric of Theorem 1 is the physically induced choice, and the sensitivity of transient conclusions to that choice is precisely the structural point of [4] on the gravitational side. The persistence of structure over vectors is quantified by the projector overlap functional $C(\Delta) = \text{Tr}(\Pi(t)\Pi(t+\Delta))$: observable channels are those with $C(\Delta) \rightarrow 1$.

6 The correspondence, at its honest grade

Theorem 3 (shared wall structure — shape side proven, GR side cited). *At the coplanar wall of Σ_4 , the linearized dynamics has the exact structure: (i) block-triangular decoupling of the transverse sector at rate $O(w_3)$; (ii) transverse oscillator frequencies $\pm i\sqrt{2\lambda}$ with amplitude suppression $1/\sqrt{2\lambda}$ and persistent phase; (iii) spectral degeneracy confined to the wall, in exact size-2 Jordan blocks under the condition of Theorem 2; (iv) contour-level data (Riesz projector, trace-log ledger) finite*

and exact through the degeneracy. Each item is a gated computation of this paper. Black-hole horizons exhibit the corresponding structure on the general-relativity side: one-way membrane behavior [13], non-normal perturbation operators with pseudospectrally unstable overtones [1], and transient dynamics governed by non-orthogonal quasinormal sums [5]. These are cited, not re-derived.

Conjecture 1 (the dictionary). *The correspondence extends level-by-level:*

<i>Shape wall $\partial\Sigma_4$ (coplanar)</i>	<i>Horizon patch $\mathcal{H}^+$</i>
<i>Jacobi fields under singular reorientation</i>	<i>static fields, proper acceleration $\rightarrow \infty$</i>
<i>mass-weighted currents</i>	<i>membrane currents [13]</i>
<i>rank-1 densities $P = NN^\top$</i>	<i>coalesced QNM projectors</i>
<i>pullback metric + trace-log ledger Q_Π</i>	<i>pseudospectral ledger of the ringdown operator</i>

No isomorphism is claimed. Establishing any row above theorem grade requires deriving the horizon-side block structure from a concrete perturbation operator (e.g. hyperboloidal Regge–Wheeler [11, 1]) and exhibiting the map — open problems 1–2 of Section 9.

Two ingredients of earlier drafts are deliberately absent from the theorem. The claimed equivalence between exceptional points and Ricci blow-up of a backreacting metric is unproven and is retired to Section 9. And the refraction law $R_1 \sin^2 \theta_1 = R_2 \sin^2 \theta_2$ is not used as a bridge: the $\sin \rightarrow \sin^2$ step is under audit across the paper series (the variational derivation in the foundational paper yields first-power Snell), and until that audit closes, no argument here rests on it.

7 Gauge extension

Internal non-Abelian charge is incorporated at the standard grade. A charge vector $q \in \mathfrak{g}$ coupled along the trajectory through $L = \frac{1}{2}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu + g_s \text{Tr}(q A_\mu)\dot{x}^\mu$ obeys the Wong equations [14, 15]

$$\dot{q} = -g_s [A_\mu \dot{x}^\mu, q], \quad \frac{d}{dt}(g_{\mu\nu}\dot{x}^\nu) - \frac{1}{2}\partial_\mu g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta = g_s \text{Tr}(q F_{\mu\nu}) \dot{x}^\nu :$$

the charge is dragged and rotated by transport, and its current back-reacts as a Lorentz-type force. This is classical [14, 15] and is used here only as the mechanism populating a gauge sector K_{gauge} .

Proposition 1 (conditional persistence of defectiveness). *Let $K = K_{\text{geom}} + K_{\text{gauge}}$ with $F_{\mu\nu} \neq 0$, and assume (i) the sectors act orthogonally, $\text{Tr}(K_{\text{geom}}^\top K_{\text{gauge}}) = 0$, and (ii) the mixed cross-commutator is strictly dominated, $\|R_{\text{mixed}}\| < g_s^2 \sum_a \|[q_i^a, q_j^a]\|^2$. Then $\|[K, K^\top]\| > 0$ even at perfect spatial isotropy $t_1 = t_2 = t_3$ (where $\|[K_{\text{geom}}, K_{\text{geom}}^\top]\| = 0$ by Section 2).*

The algebra is a reverse triangle inequality and is elementary; the content lives entirely in hypothesis (ii), which asserts rather than proves that the gauge sector’s non-normality is not cancelled by cross terms. The proposition is therefore *conditional*, and establishing (ii) for a concrete field configuration is open problem 4.

8 Speculative extensions, fenced

Nothing in this section is derived; each item is an ansatz or analogy, recorded because it points somewhere concrete.

Superradiant extraction rate. Motivated by the established transient superradiance of [6], an extraction-rate ansatz for the gauge-extended shape system is

$$\frac{dE}{dt} \stackrel{?}{=} \alpha_s \int_{\Sigma_4} \text{Tr}(J_{\text{color}}^\mu F_{\mu\nu} \dot{W}^\nu) \|J\|^2 \sqrt{|g|} d^5w + \gamma \int_{\Sigma_4} \text{Tr}([D_t K, K^\top](J J^\dagger)) \sqrt{|g|} d^5w,$$

an alignment term plus an advective pseudospectral driver. Its derivation from the $H^1(\Sigma_4)$ energy functional, with the quotient metric of Theorem 1 in place, has not been carried out.

Junction matching. A Darboux–Israel-type matching [18] of the discrete wall data to a continuum exterior, sourcing a surface stress from the lattice turning angle, is a proposed program, not a result: the bridge identity equating the extrinsic-curvature jump with lattice parameters is an ansatz.

Condensed-matter analogue. Polariton interfaces with non-Hermitian mass-coupling matrices plausibly realize the same rigidity-interface structure near their exceptional points; a quantitative trapping law awaits a dimensionally consistent derivation (the closed formula of earlier drafts is withdrawn as heuristic).

9 Open problems

1. Derive the horizon-side block structure of Theorem 3 from a concrete non-selfadjoint ringdown operator on hyperboloidal slices [1], and exhibit the map of Conjecture 1 row by row.
2. Compute the Bures–Wasserstein/qutrit dictionary’s physical content: what does the quantum-information structure of Σ_4 (fidelity, quantum Fisher information) measure mechanically?
3. Close the sin vs. $\sin^2$ audit of the refraction law across the series; only then may it re-enter the correspondence.
4. Establish or refute hypothesis (ii) of Proposition 1 for a concrete non-Abelian background.
5. The exceptional-point/Ricci-blow-up equivalence of earlier drafts: prove it for the backreacting metric ansatz, or retire it permanently.
6. Extract the closed forms of the wall-curvature constants ($K_{44}w_3 \rightarrow 0.0713\dots$, $K_{55}w_3 \rightarrow 0.1033\dots$, $K_{22}w_3 \rightarrow 0.701\dots$) from the O’Neill formula analytically.
7. The square-case ladder: the same construction at $d = 4, 5$ ($N = 5, 6$ bodies) gives Bures geometries of higher-rank state spaces; which wall results survive?

A Projector persistence bounds

Let A be a bounded operator, Γ a closed contour isolating $\sigma_\Gamma(A) \subset \sigma(A)$ with spectral gap $\delta_{\text{gap}} = \text{dist}(\Gamma, \sigma(A) \setminus \sigma_\Gamma(A)) > 0$, and $A' = A + E$, $\|E\| \leq \epsilon$. With $R_A(\lambda) = (\lambda I - A)^{-1}$ and the pseudospectral growth factor $C_\Gamma = \sup_{\lambda \in \Gamma} \|R_A(\lambda)\| \cdot \delta_{\text{gap}}$:

Lemma 1 (Neumann resolvent bound). *If $\epsilon < (\sup_{\lambda \in \Gamma} \|R_A(\lambda)\|)^{-1}$ then $R_{A'}(\lambda)$ exists on Γ with $\|R_{A'}(\lambda)\| \leq \|R_A(\lambda)\|/(1 - \|R_A(\lambda)\| \epsilon)$.*

Proof. $\lambda I - A' = (\lambda I - A)(I - R_A(\lambda)E)$; the Neumann series for the second factor converges geometrically under the hypothesis, and the norm bound is the geometric sum. $\square$

Theorem 4 (projector persistence Lipschitz bound). *For $\epsilon < \delta_{\text{gap}}/C_\Gamma$, the Riesz projectors Π, Π' over Γ satisfy*

$$\|\Pi' - \Pi\| \leq \frac{\ell(\Gamma) C_\Gamma^2}{2\pi \delta_{\text{gap}} (\delta_{\text{gap}} - \epsilon C_\Gamma)} \epsilon,$$

so the map from operators to their Grassmannian projector data is locally Lipschitz: higher-rank spectral subspaces vary continuously even where individual eigenvectors are defective.

Proof. $\Pi' - \Pi = -\frac{1}{2\pi i} \oint_\Gamma R_A(\lambda) E R_{A'}(\lambda) d\lambda$ by the resolvent identity; apply the lemma and the triangle inequality for integrals. $\square$

These are standard Kato-type estimates [10, 8]; they are included because Section 4’s demonstration realizes their hypotheses concretely — including the failure mode when the contour does not enclose the colliding pair.

B The scripts: proof of record

Both scripts run standalone (Python 3.12.3, NumPy, mpmath 1.3.0) from a clean directory. **BH-CURV-01** carries six gates: the Kendall $d = 2$ control (sectional curvature 4.0000000), quadratic-form symmetry (exact), $R(V, \cdot)V \cdot V = 0$ (4×10^{-11}), the closed-form quotient metric (Theorem 1), interior finiteness at $w_1 \rightarrow w_2$, and the wall laws $g_{22} \cdot 6w_3 \rightarrow 1.000101$ with curvature $\sim C/w_3$. **BH-LEDGER-01** carries three: the snapped divergence law $\|P_1\| \cdot \text{gap} \rightarrow |A_{12}|$, $\text{Tr } \Pi = 2.000000000000$ through and at the Jordan point, and the trace-log ledger exact to 6×10^{-15} at the defective point.

BH-CURV-01 — quotient metric and O’Neill curvature of Σ_4 ; six gates

```

1 # SCRIPT: BH-CURV-01 -- fresh derivation of the Jacobi curvature term on Sigma_4
2 # via O’Neill for the submersion S^{d^2-1} (flat sphere of X’s, ||X||_F=1) -> Sigma
3 # under X -> RX (SO(d) left action), base coordinate G = X^T X (the audited invariant).
4 # GATES:
5 # G1: d=2 control -- Kendall shape sphere sectional curvature == 4 (exact known value)
6 # G2: symmetry K_curv(a,b) = K_curv(b,a)
7 # G3: row/column along V vanishes (R(V,V)V structure)
8 # G4: quotient metric g_ab = <H_a, H_b> reported (tests the paper’s silent g=I assumption)
9 # G5: blow-up structure of shear entries as w1 -> w2 (exponent + which entries)
10 import mpmath as mp
11 import numpy as np
12 mp.mp.dps = 40
13
14 def frob(A,B): return float('nan') # placeholder (numpy used below)
15
16 def make_basis3():
17     import numpy as np
18     E1 = np.diag([1,-1,0])/np.sqrt(2)
19     E2 = np.diag([1,1,-2])/np.sqrt(6)
20     E3 = np.zeros((3,3)); E3[0,1]=E3[1,0]=1/np.sqrt(2)
21     E4 = np.zeros((3,3)); E4[0,2]=E4[2,0]=1/np.sqrt(2)
22     E5 = np.zeros((3,3)); E5[1,2]=E5[2,1]=1/np.sqrt(2)
23     return [E1,E2,E3,E4,E5]
24
25 def skew_basis(d):
26     import numpy as np
27     out=[]
28     for i in range(d):
29         for j in range(i+1,d):

```

```

30         O=np.zeros((d,d)); O[i,j]=1; O[j,i]=-1
31         out.append(O)
32     return out
33
34 class Submersion:
35     def __init__(self,d):
36         import numpy as np
37         self.d=d; self.np=np
38         self.sk=skew_basis(d)
39     def lift(self,X,E):
40         # solve  $X^T H + H^T X = E$  (sym,  $d(d+1)/2$  eqs),  $X H^T - H X^T = 0$  (skew,  $d(d-1)/2$  eqs)
41         np=self.np; d=self.d
42         rows=[]; rhs=[]
43         def vec(i,j):
44             z=np.zeros((d,d)); z[i,j]=1; return z
45         cols=[(i,j) for i in range(d) for j in range(d)]
46         Msys=np.zeros((d*d,d*d))
47         b=np.zeros(d*d)
48         r=0
49         for i in range(d):
50             for j in range(i,d):
51                 for c,(k,l) in enumerate(cols):
52                     Hc=vec(k,l)
53                     val=(X.T@Hc + Hc.T@X)[i,j]
54                     Msys[r,c]=val
55                     b[r]=E[i,j]; r+=1
56         for i in range(d):
57             for j in range(i+1,d):
58                 for c,(k,l) in enumerate(cols):
59                     Hc=vec(k,l)
60                     val=(X@Hc.T - Hc@X.T)[i,j]
61                     Msys[r,c]=val
62                     b[r]=0.0; r+=1
63         h=np.linalg.solve(Msys,b)
64         return h.reshape(d,d)
65     def sphere_proj(self,X,Z):
66         return Z - (self.np.sum(X*Z))*X
67     def vert_proj(self,X,Z):
68         np=self.np
69         Vb=[0@X for 0 in self.sk]
70         G=np.array([[np.sum(a*b) for b in Vb] for a in Vb])
71         rhs=np.array([np.sum(a*Z) for a in Vb])
72         c=np.linalg.solve(G,rhs)
73         V=sum(ci*vi for ci,vi in zip(c,Vb))
74         return V
75     def A_tensor(self,X,H1,E2,eps=1e-6):
76         #  $A_{\{H1\}}$  (lift field of  $E2$ ) = vert( sphere-covariant derivative along  $H1$  )
77         np=self.np
78         Xp=X+eps*H1; Xp=Xp/np.linalg.norm(Xp)
79         Xm=X-eps*H1; Xm=Xm/np.linalg.norm(Xm)
80         Lp=self.lift(Xp,E2); Lm=self.lift(Xm,E2)
81         D=(Lp-Lm)/(2*eps)
82         D=self.sphere_proj(X,D)
83         return self.vert_proj(X,D)
84
85 def sigma4_curvature(w, Vcoef, eps=1e-6):
86     import numpy as np
87     d=3; sub=Submersion(3)
88     s=np.sqrt(np.array(w)); X=np.diag(s)
89     Es=make_basis3()
90     Hs=[sub.lift(X,E) for E in Es]
91     g=np.array([[np.sum(Hs[a]*Hs[b]) for b in range(5)] for a in range(5)])
92     HV=sum(c*H for c,H in zip(Vcoef,Hs))
93     AV=[sub.A_tensor(X,HV,Es[b],eps) for b in range(5)]
94     K=np.zeros((5,5))
95     gV=np.array([np.sum(HV*Hs[a]) for a in range(5)]); VV=np.sum(HV*HV)
96     for a in range(5):
97         for b in range(5):

```

```

98         sph=VV*g[a,b]-gV[a]*gV[b]
99         K[a,b]=sph+3*np.sum(AV[a]*AV[b])
100     return K,g
101
102 import numpy as np
103 print("=== GATE G1: Kendall d=2 control (expect sectional curvature 4) ===")
104 sub2=Submersion(2)
105 for (a,b) in [(0.7,0.3),(0.5,0.5),(0.9,0.1)]:
106     X=np.diag([np.sqrt(a),np.sqrt(b)])
107     # base tangent basis on trace-one symmetric 2x2: diag direction + offdiag
108     F1=np.diag([1,-1])/np.sqrt(2)
109     F2=np.array([[0,1],[1,0]])/np.sqrt(2)
110     H1=sub2.lift(X,F1); H2=sub2.lift(X,F2)
111     g11=np.sum(H1*H1); g22=np.sum(H2*H2); g12=np.sum(H1*H2)
112     A12=sub2.A_tensor(X,H1,F2)
113     num=(1.0*(g22*g11-g12**2))+3*np.sum(A12*A12) # sphere K=1 sectional part + 0'Neill
114     K_sec=num/(g11*g22-g12**2)
115     print(f" w=({a},{b}): sectional K = {K_sec:.10f}")
116
117 print("\n=== Sigma_4: curvature quadratic form K_curv(a,b) for generic V ===")
118 w=(0.5,0.3,0.2)
119 V=np.array([0.3,-0.2,0.4,0.1,-0.5]); V=V/np.linalg.norm(V)
120 K,g=sigma4_curvature(w,V)
121 print("quotient metric g_ab (GATE G4 -- is it identity?):")
122 print(np.array2string(g,precision=6,suppress_small=True))
123 print("K_curv:")
124 print(np.array2string(K,precision=6,suppress_small=True))
125 print("GATE G2 symmetry: max|K-K^T| =",np.max(np.abs(K-K.T)))
126 KV=K@np.linalg.solve(g,np.array([np.sum(V[a]*g[a]) for a in range(5)])) if False else None
127 # G3: contract with V in the metric sense: K acting paired with V-lift should vanish
128 gV=g@V
129 print("GATE G3 |K . (g^{-1} gV)| structure: |K @ V| =",np.max(np.abs(K@np.linalg.solve(g,gV))))
130
131 print("\n=== GATE G5: blow-up as w1 -> w2 (V purely in shape direction E1) ===")
132 Vs=np.array([1.0,0,0,0,0])
133 for dlt in [1e-1,1e-2,1e-3,1e-4]:
134     w=(0.4+dlt/2,0.4-dlt/2,0.2)
135     K,g=sigma4_curvature(w,Vs,eps=min(1e-6,dlt*1e-3))
136     print(f" w1-w2={dlt:.0e}: K33={K[2,2]:.6e} K44={K[3,3]:.6e} K55={K[4,4]:.6e} g33={g[2,2]:.6e}")
137
138 print("\n=== GATE G6: the TRUE wall -- blow-up at the coplanar stratum w3 -> 0 ===")
139 print("metric prediction: g22 = sum (E2)_ii^2/(4 w_i) contains 4/(24 w3) = 1/(6 w3) -> infinity")
140 Vg=np.array([0.3,-0.2,0.4,0.1,-0.5]); Vg=Vg/np.linalg.norm(Vg)
141 for w3 in [1e-1,1e-2,1e-3,1e-4]:
142     w=(0.55,0.45-0, w3); w=(0.55*(1-w3)/1.0, 0.45*(1-w3)/1.0, w3)
143     K,g=sigma4_curvature(w,Vg,eps=min(1e-6,w3*1e-2))
144     pred=1/(6*w3)
145     print(f" w3={w3:.0e}: g22={g[1,1]:.6e} (1/(6w3)={pred:.3e}) K44={K[3,3]:.4e} K55={K[4,4]:.4e} K22={K[1,1]:.4e}")
146 print("fit: K44*w3, K55*w3 ->")
147 vals=[]
148 for w3 in [1e-2,1e-3,1e-4]:
149     w=(0.55*(1-w3), 0.45*(1-w3), w3)
150     K,g=sigma4_curvature(w,Vg,eps=min(1e-6,w3*1e-2))
151     vals.append((w3,K[3,3]*w3,K[4,4]*w3,g[1,1]*6*w3))
152 for v in vals: print(f" w3={v[0]:.0e}: K44*w3={v[1]:.6f} K55*w3={v[2]:.6f} g22*6*w3={v[3]:.6f}")

```

BH-LEDGER-01 — the Dunford ledger at the wall's Jordan point; three gates

```

1 # SCRIPT: BH-LEDGER-01 -- the Dunford trace-log ledger, computed on the audited operator
2 # at the coplanar wall's Jordan point. His tip made precise:
3 # rank-one (eigenvector) data DIVERGES ~ 1/gap approaching the Jordan point;
4 # contour (Dunford/Riesz) data stays EXACT, including AT the defective point.
5 # Audited shape block (four-body audit, FOURBODY-AUDIT scripts):
6 # A11 = (t1+t2) - (t1-t2)(w1-w2); A22 = 2 t3 + w3 (t1+t2-2t3)
7 # A12 = [ (t1-t2) - (t1+t2-2t3)(w1-w2) ] / sqrt(3); A21 = sqrt(3) (t1-t2) w3

```

```

8 # Shear block diagonal: t1+t2, t1+t3, t2+t3 (separated spectators).
9 # Jordan condition at w3=0: S = a (w1-w2), S = t1+t2-2t3, a = t1-t2.
10 import mpmath as mp
11 mp.mp.dps = 30
12 import numpy as np
13
14 t1,t2,t3 = 2.0, 1.0, 1.3 # a = 1, S = 0.4
15 dw = 0.4 # w1 - w2 = S/a -> Jordan at w3 = 0
16 lamstar = 2*t3 # = A11 = A22 at the Jordan point = 2.6
17
18 def block(eps):
19     w1 = 0.7 - eps/2; w2 = 0.3 - eps/2; w3 = eps
20     A11 = (t1+t2) - (t1-t2)*(w1-w2)
21     A22 = 2*t3 + w3*(t1+t2-2*t3)
22     A12 = ((t1-t2) - (t1+t2-2*t3)*(w1-w2))/np.sqrt(3)
23     A21 = np.sqrt(3)*(t1-t2)*w3
24     return np.array([[A11,A12],[A21,A22]])
25
26 def riesz(A, z0, r, npts=4000, f=None):
27     # (1/2pi) contour-int f(lambda) (lambdaI - A)^{-1} dlambda over circle center z0 radius r
28     n = A.shape[0]
29     acc = np.zeros((n,n), dtype=complex)
30     for k in range(npts):
31         th = 2*np.pi*k/npts
32         lam = z0 + r*np.exp(1j*th)
33         R = np.linalg.inv(lam*np.eye(n) - A)
34         w = (f(lam) if f else 1.0)
35         acc += w * R * (1j*r*np.exp(1j*th)) * (2*np.pi/npts)
36     return acc/(2j*np.pi)
37
38 print("=== approach along w3 -> 0+ : rank-one data vs contour data ===")
39 print(f"{'eps':>8} {'gap':>12} {'||P1|| rank-one':>16} {'Tr Pi (contour)':>18} {'||Pi_blk||':>12} {'ledger - 2 log(lam*):>20}")
40 for eps in [1e-1,1e-2,1e-3,1e-4,1e-6,0.0]:
41     A = block(eps)
42     ev, V = np.linalg.eig(A)
43     gap = abs(ev[0]-ev[1])
44     if gap > 1e-14:
45         Vi = np.linalg.inv(V)
46         P1 = np.outer(V[:,0], Vi[0,:])
47         p1n = np.linalg.norm(P1)
48     else:
49         p1n = float('inf')
50     z0 = np.trace(A)/2
51     Pi = riesz(A, z0, 0.15)
52     led = riesz(A, z0, 0.15, f=lambda z: np.log(z))
53     ledger = np.trace(led).real
54     target = 2*np.log(np.trace(A).real/2)
55     print(f"{'eps':>8.0e} {'gap':>12.3e} {'p1n':>16.4e} {'np.trace(Pi).real:>18.12f} {'np.linalg.norm(Pi):>12.6f} { ledger-target:>20.3e}")
56
57 print("\n[GATE L1] rank-one projector norm diverges ~ 1/gap: compare ||P1||*gap:")
58 for eps in [1e-2,1e-3,1e-4]:
59     A = block(eps); ev,V = np.linalg.eig(A)
60     gap = abs(ev[0]-ev[1]); Vi = np.linalg.inv(V)
61     P1 = np.outer(V[:,0],Vi[0,:])
62     print(f"eps={eps:.0e}: ||P1||*gap = {np.linalg.norm(P1)*gap:.6f}")
63 A0 = block(0.0)
64 print(f"\n[GATE L2] AT the Jordan point (eps=0): A = {A0.tolist()}")
65 print(f" off-diagonal A12 = {A0[0,1]:.6f} (nonzero -> genuinely defective), A21 = {A0[1,0]:.1f}")
66 Pi0 = riesz(A0, lamstar, 0.15)
67 led0 = riesz(A0, lamstar, 0.15, f=lambda z: np.log(z))
68 print(f" Tr Pi (contour) = {np.trace(Pi0).real:.12f} (exact: 2)")
69 print(f" ||Pi_blk|| = {np.linalg.norm(Pi0):.6f} (bounded; = ||I2|| + nilpotent part)")
70 print(f" trace-log ledger = {np.trace(led0).real:.12f} vs 2 log(lam*) = {2*np.log(lamstar):.12f}")
71 print(f"[GATE L3] ledger exact at the defective point: {abs(np.trace(led0).real-2*np.log(lamstar)) < 1e-9}")

```

C Graphical summary



The Hairdresser toolkit, with every panel naming a gated computation: the scissors ($A_{21} = O(w_3) \rightarrow 0$, block-triangular at the wall), the fingers (transverse amplitude $\propto 1/\sqrt{2\lambda}$, phase persists), the mirror (trace-log ledger exact at the defective point), and the light ($\operatorname{Tr} \Pi = 2$ through the collision; rank-one data diverges as $|A_{12}|/\text{gap}$).

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